

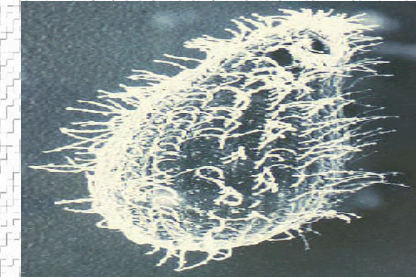
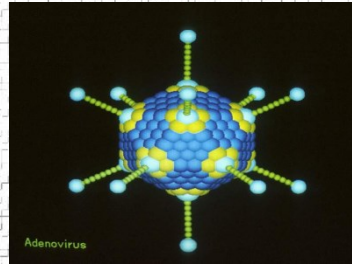
Module 1: Introduction to Medical Microbiology

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MEDICAL MICROBIOLOGY



= “the study of **microorganisms** that **affect health**”

= “... study of ... **interactions** between animals (primarily **humans**) and **microorganisms** of **medical importance**...”

MEDICAL MICROBIOLOGIST

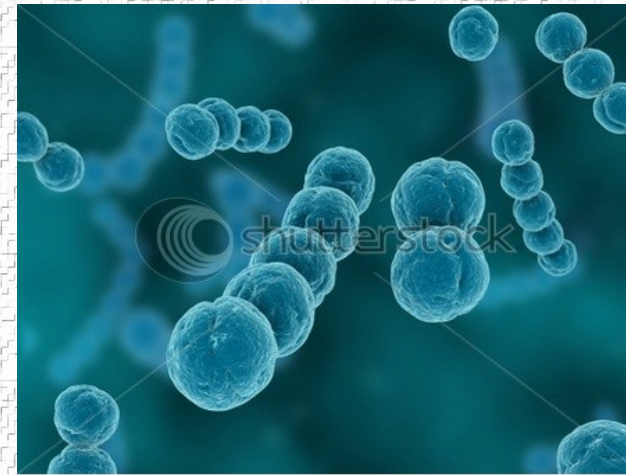
= studies the following about microbial pathogens :

characteristics

growth

mode(s) of transmission

mechanisms of infection, etc.



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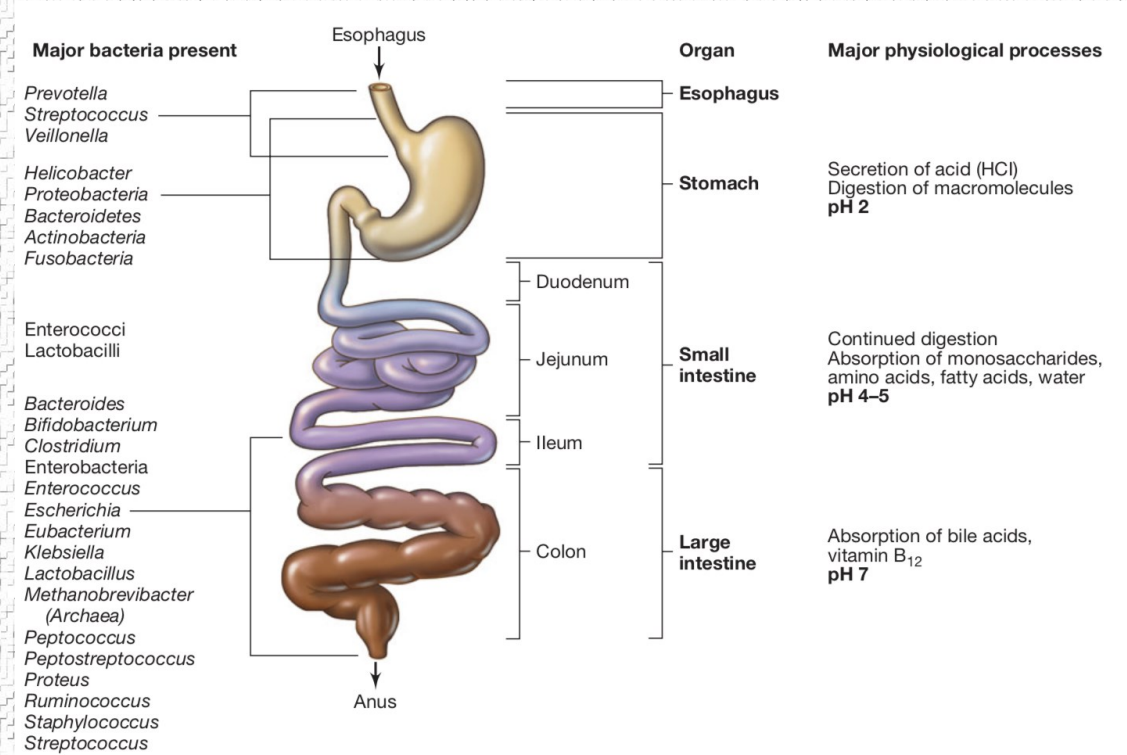
OUTCOME OF THE INTERACTION(S)

1. Beneficial

= may contribute to the survival or to the over-all health of the individual;

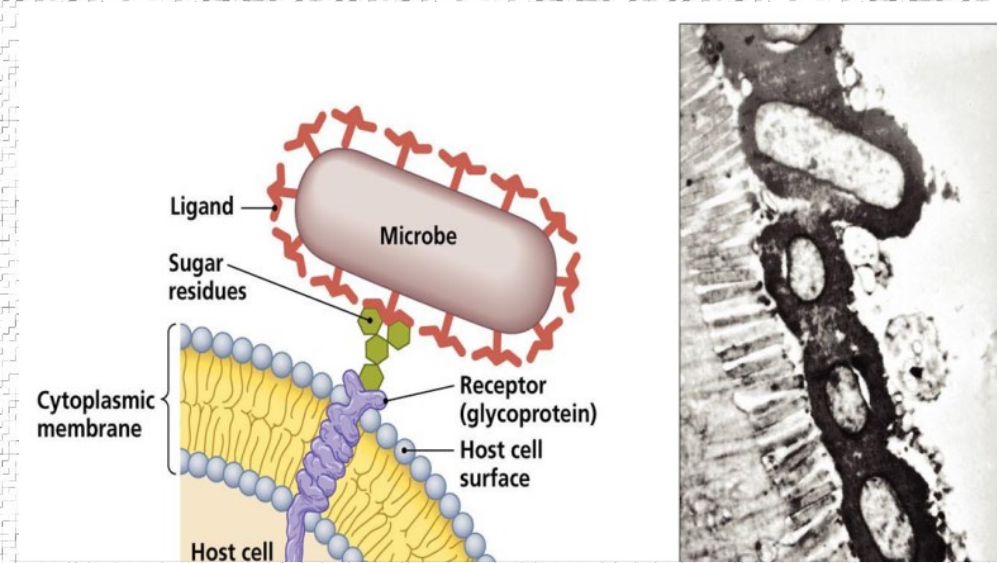
= *particularly true for the **endogenous or normal microbial flora** (if present at normal levels)* ; if levels increase, it can cause diseases from opportunistic microorganisms

a• **participate in the metabolism of food products and aid in the absorption of nutrients**



b• **provide essential growth factors (e.g. Vit. K, thiamine, riboflavin, pyridoxine, etc.)**

c. protect against infections with highly virulent microorganisms (such as through prevention of colonization by bacterial interference)



- d. competition for nutrients and for receptor sites on the host cell**
- e. production of bacteriocins and other inhibitory substances**
- f. stimulate the immune response, etc. ; creates antibodies**

2. Harmful (may lead to disease)

Disease = “ an **abnormal state ...
a part or all of the body is
not properly adjusted or is
not capable of carrying out
its normal functions”**



**= “ an injury to the host that
impairs host function”**

Two Groups of Microorganisms Associated with the Human Body

1. The Transient Flora
2. The Resident Flora



<https://www.pastiche-training.com/flosaid/the-importance-of-the-skin-surface-microflora>

THE TRANSIENT FLORA

- **environment-derived microorganisms**
- **do not inhabit body tissues permanently** (only hours, days or weeks)
- **generally, not significant in the presence of an intact normal microflora (NMF)**

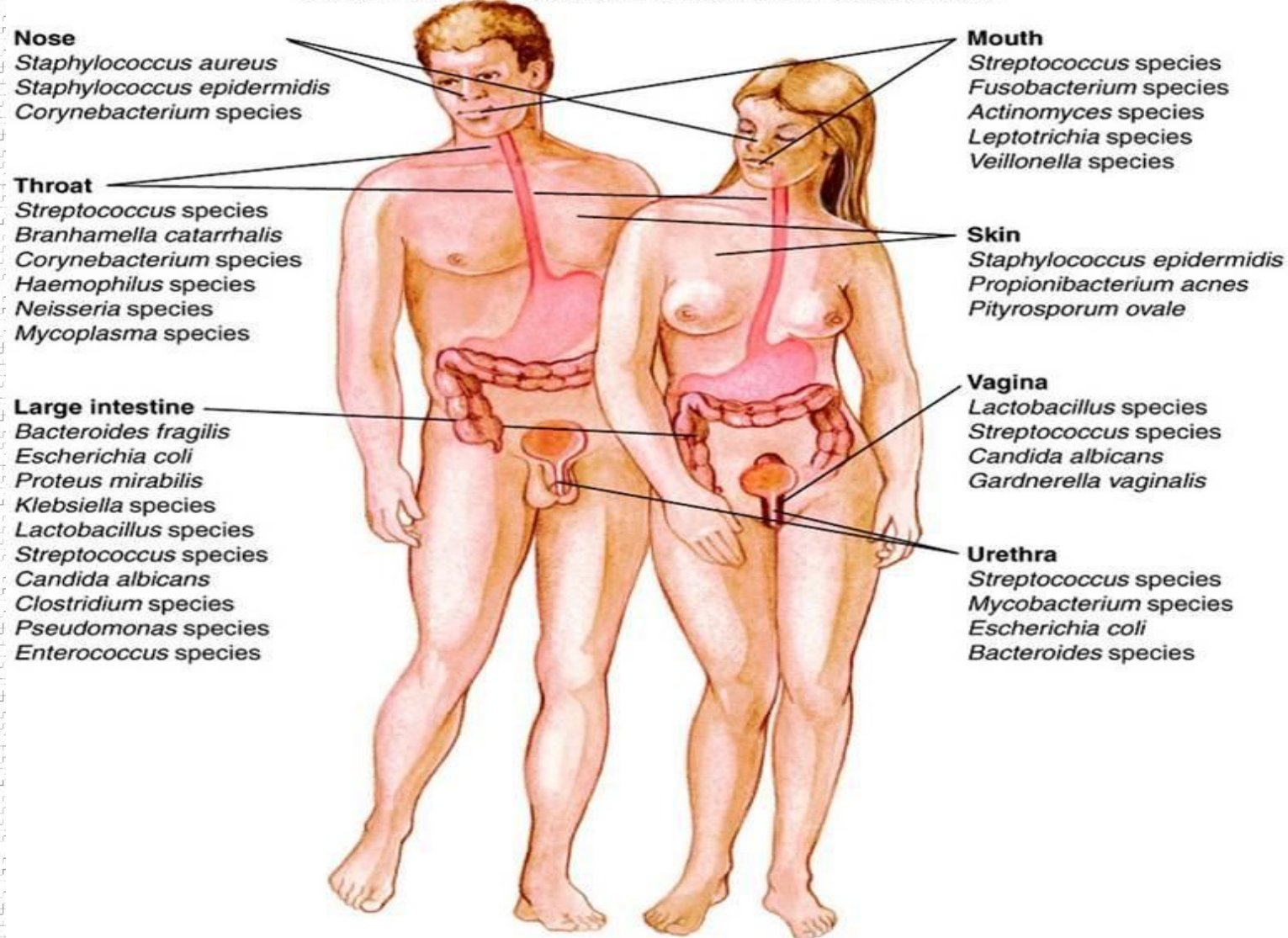
THE RESIDENT OR NORMAL MICROFLORA OF THE HUMAN BODY

- “ the population of microorganisms that inhabit the **skin** and **mucous membranes** (MM) of healthy, normal persons”
- a collective term for “species that have developed an intimate relationship with certain tissues of the animal body”

Every individual has a different NMF due to environmental and internal factors.

THE RESIDENT OR NORMAL MICROFLORA OF THE HUMAN BODY

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**MM = ... tightly packed epithelial
cells (single- or multi-layered),
... frequently covered with mucus;
... exist in direct contact with
the external environment**

❖ Where are MM found?

THE RESIDENT FLORA OR NORMAL MF

- **generally ~ fixed types, commensals**
- **regularly associated in a given area of the body and at a specified age**
- **promptly re-establish themselves when “disturbed” (e.g. during use of antibiotics)**

THE RESIDENT FLORA :

- normally harmless in their **regular location** in the host and in the **absence of “coincident abnormalities”** (e.g. immunodeficiency)

THE RESIDENT FLORA :

- may cause disease if :

→ introduced into “foreign” areas in the host; and

→ in the presence of predisposing factors which are behaviors related to a particular disease (e.g. sedentary lifestyle = heart diseases)

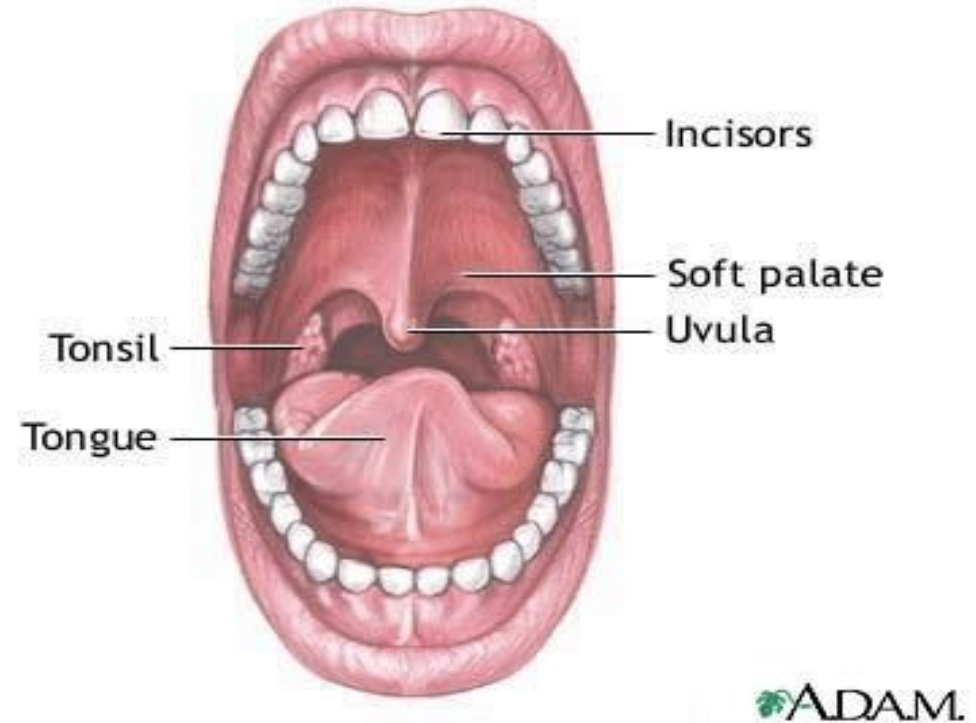
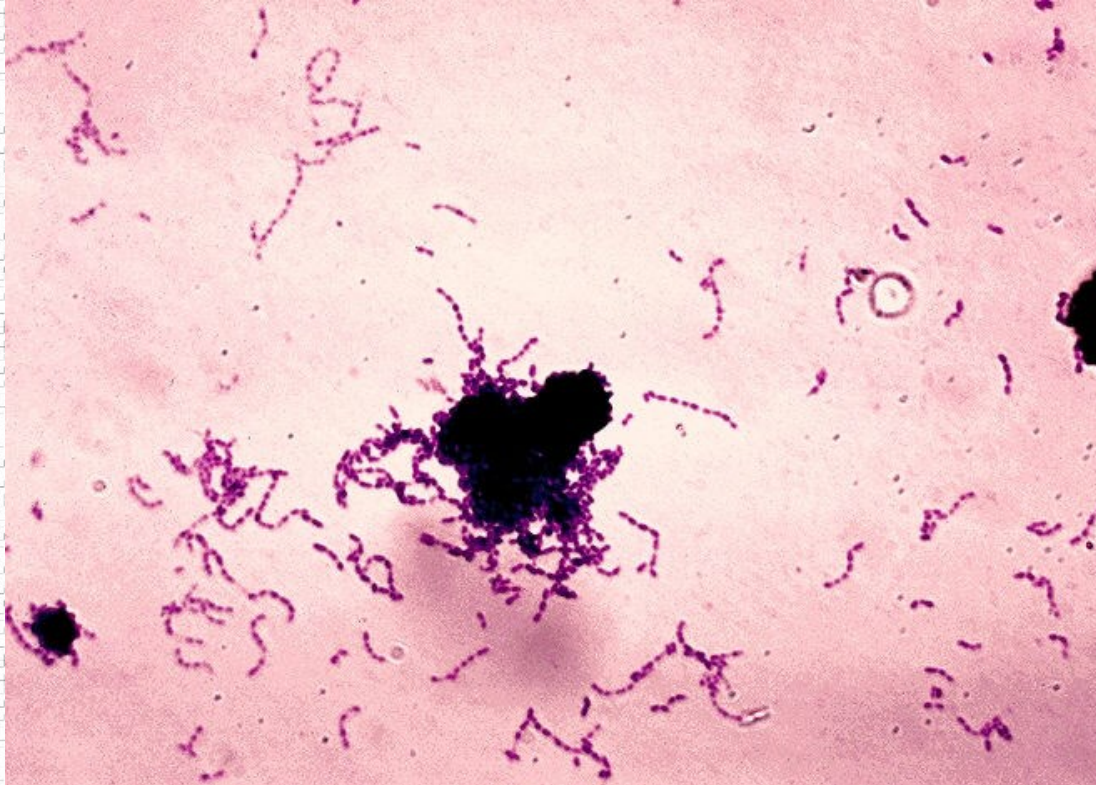
- **Introduction to “foreign” areas in the host : examples**

1. **Viridans streptococci (NMF of URT)**

2. ***Bacteroides* (NMF of colon)**

Viridans Streptococci (NMF of Mouth and URT)

- partial hemolysis of hemoglobin in the RBCs
- endocarditis = inflammation of the heart

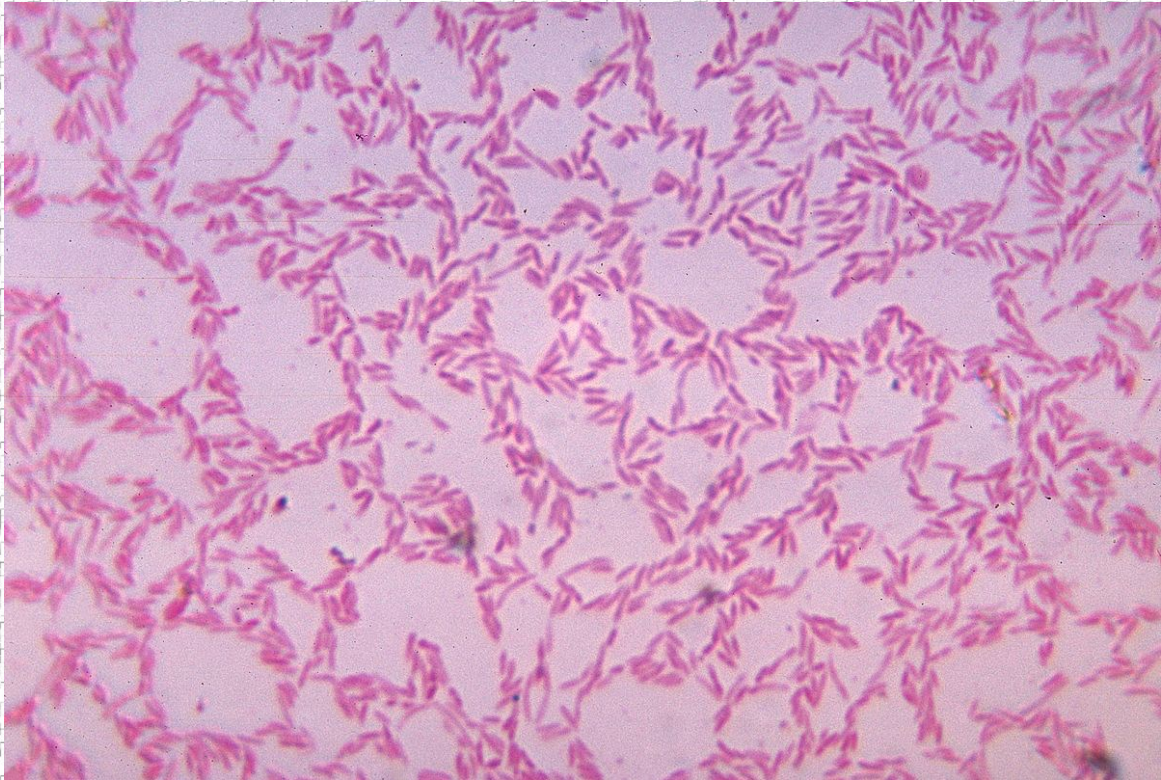


e.g. *S. mutans*

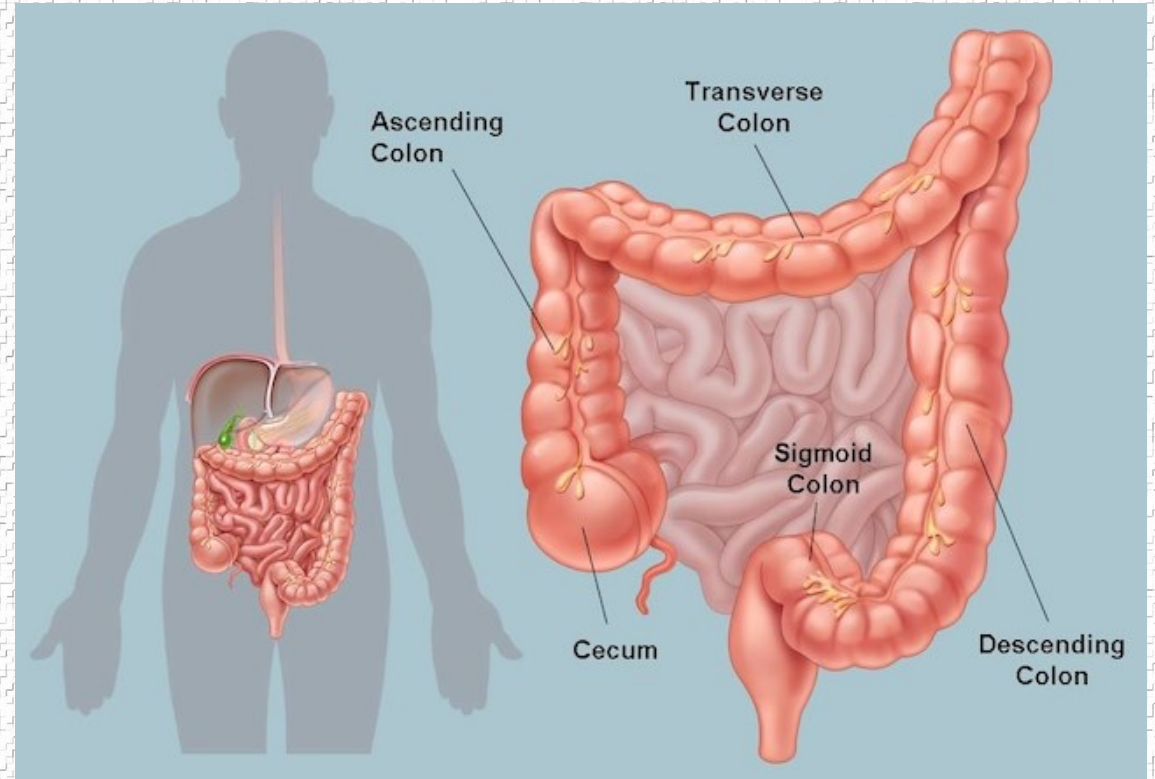
https://en.wikipedia.org/wiki/Streptococcus_mutans

***Bacteroides* (NMF of colon)**

- gram-negative, rod-shaped, anaerobic, commensal/mutualistic
- introduction to the stomach leads to stomach ulcer



<https://en.wikipedia.org/wiki/Bacteroides>



<https://www.webmd.com/digestive-disorders/picture-of-the-colon>

- **examples of predisposing factors :**

- 1. oral contraceptive use (OC)**

**→ may disrupt the normal acidity
and moisture of GIT**

- 2. diabetes**

**→ hyperglycemia favors immune
system dysfunction**

NMF :

- **not usually essential for life ... but these may help maintain health & normal function**
- **not normally found in ...the blood, lymph & many other internal organs**
- **knowledge of these is important ... as it may influence the perception of whether there is disease or not**

How did we get colonized?

Known Facts :

- Animals (including humans) are generally “germ-free” *in utero* –has been challenged by researches in the past few years

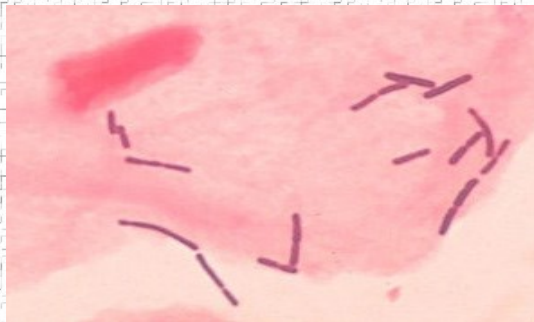
* except in some cases ...
(e.g. when infected by “**TORCHES**”)

Toxoplasma gondii
Other viruses (HIV, measles)
Rubella (German measles)
Cytomegalovirus
Herpes simplex

- The development and establishment of “normal and characteristic” microbial populations start at the time of birth. *



***Lactobacilli**



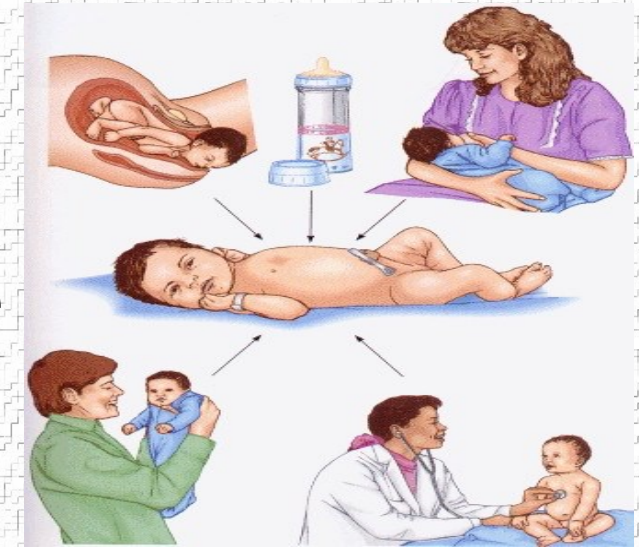
<http://www.losmicrobios.com.ar/microbios/imagenes/>

= ~ the first to establish contact with the newborn

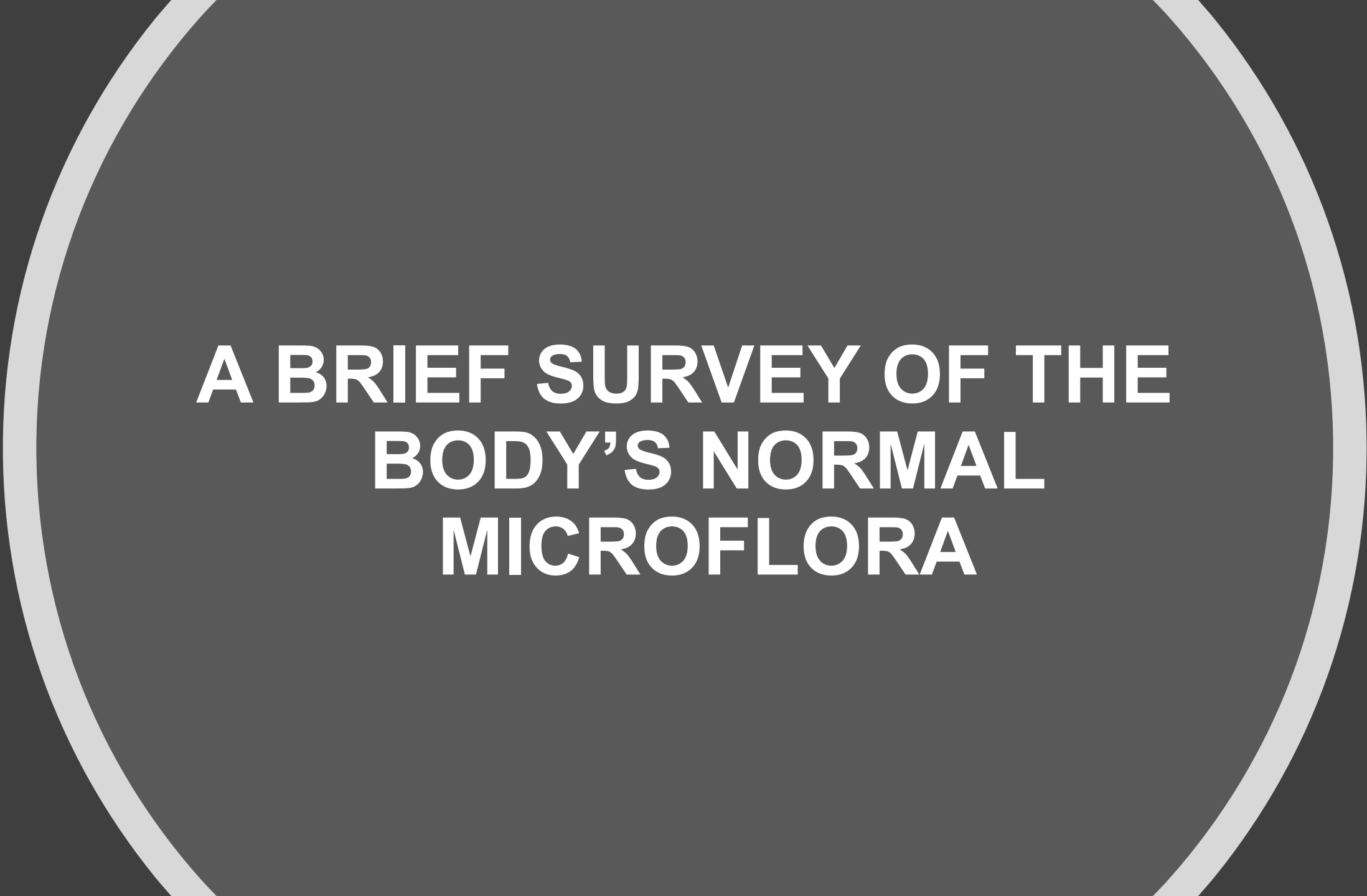


predominate in the intestine

- More microorganisms are then introduced during : **breathing and feeding events**
- Other harmless microbes colonize the other parts of the body **exposed to the external environment.**

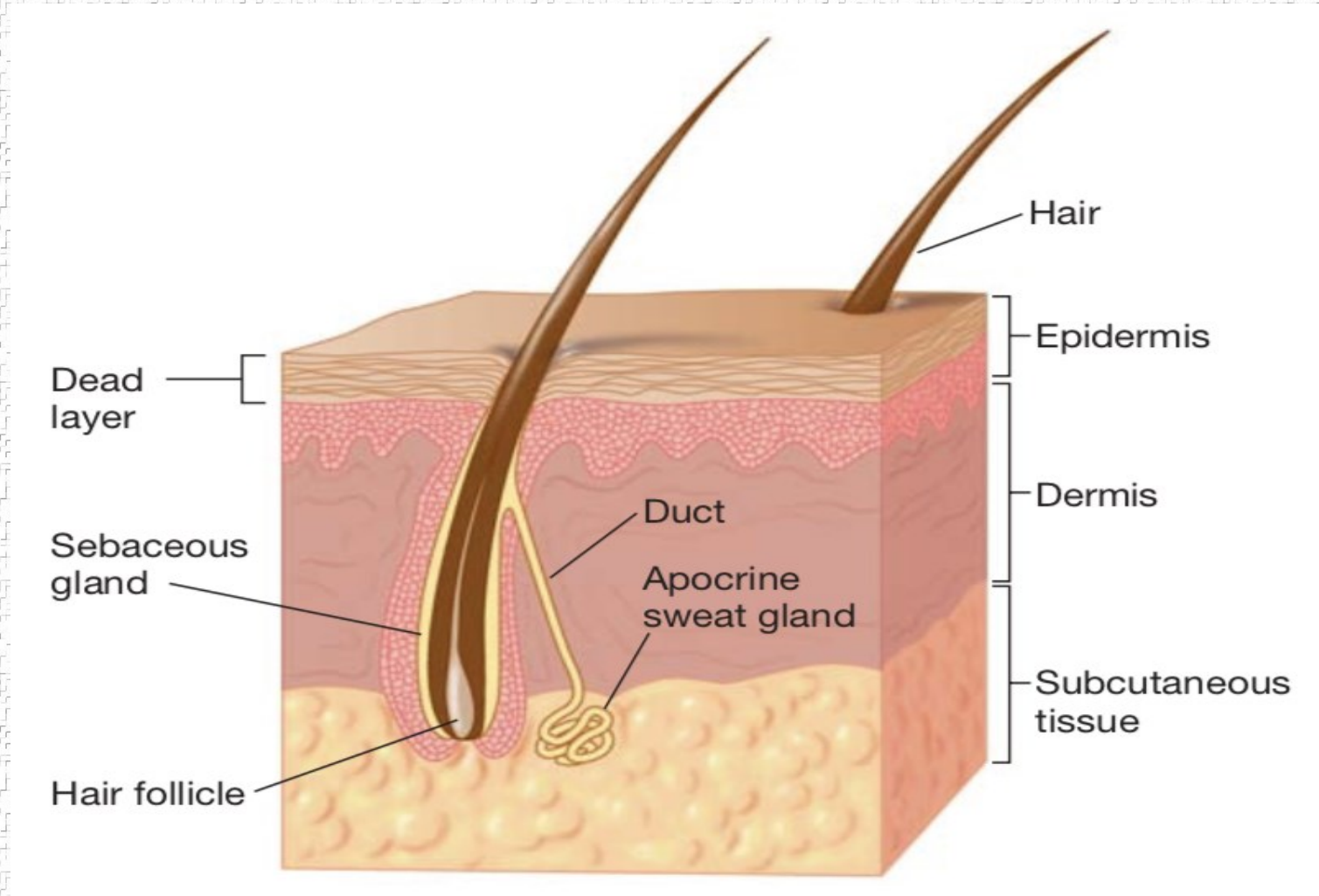


Skin and MM



A BRIEF SURVEY OF THE BODY'S NORMAL MICROFLORA

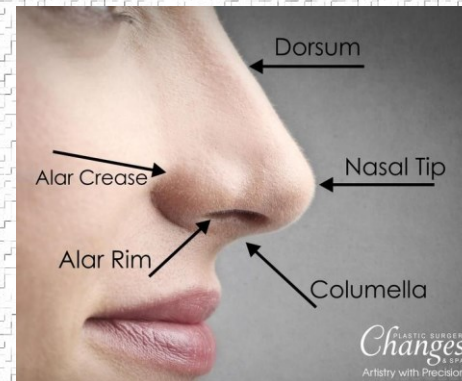
SKIN

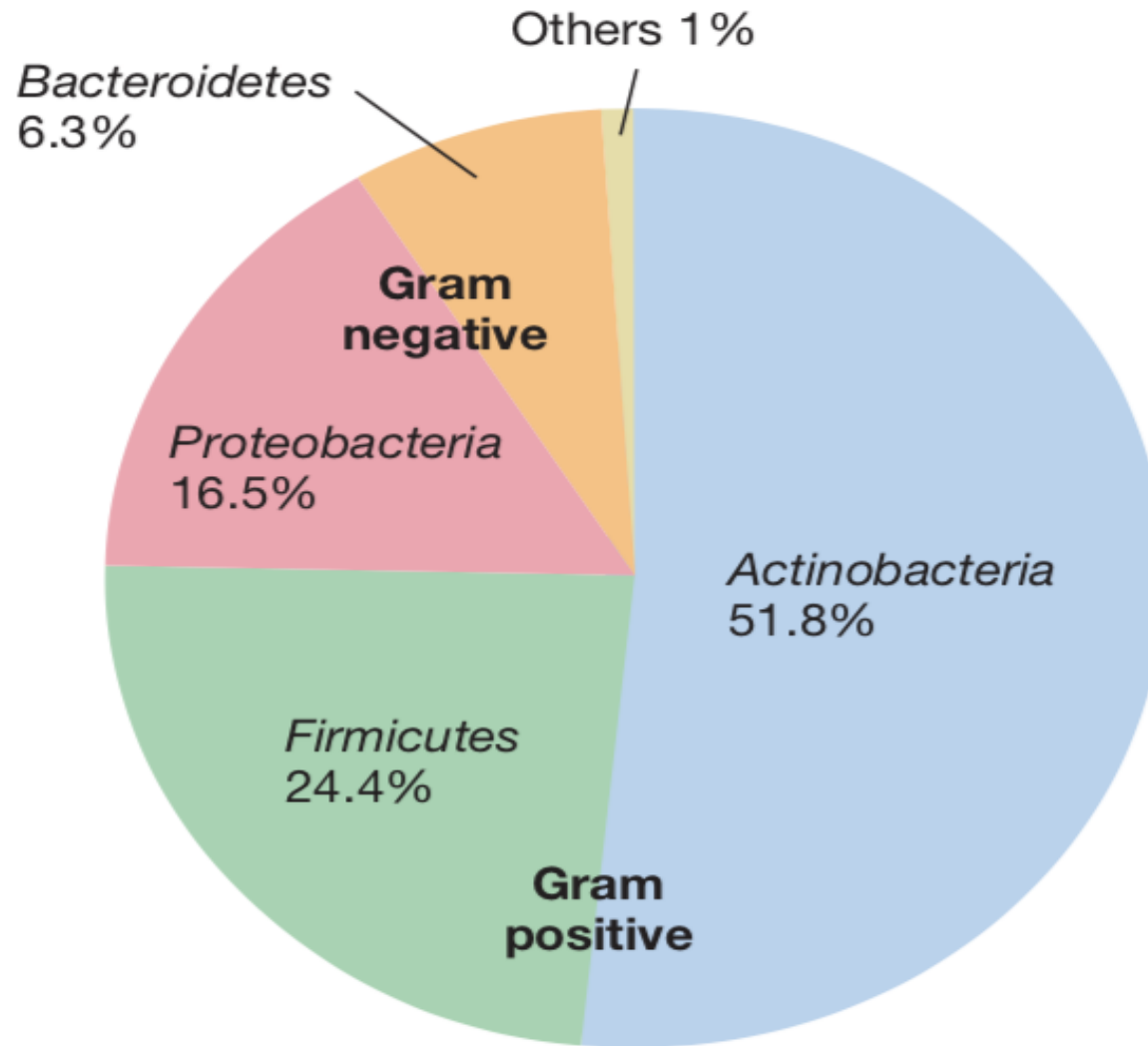


- An average adult human has about 2 m² of skin surface that varies greatly in chemical composition and moisture content.

3 Microenvironments present in the skin

1. **Moist skin areas** (e.g. inside of nostril, armpit, umbilicus etc.).
2. **Dry skin areas** (e.g. forearms and palms).
3. **Sebaceous skin** (e.g. alar crease, back of scalp, upper chest and back)

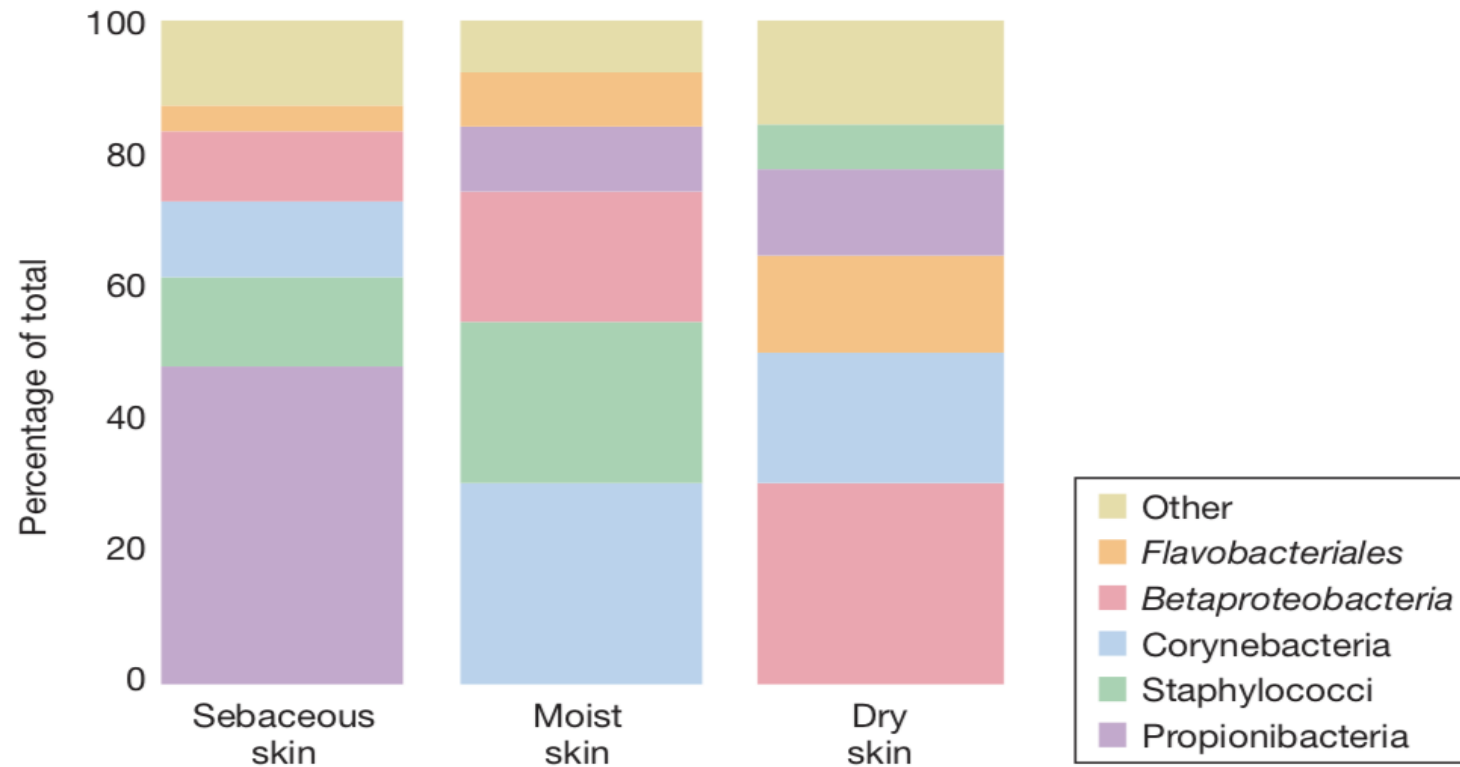




Normal Skin Microflora

(data adapted from: Grice et al., 2009, *Science* 324: 1190)

NMF of the Skin: Bacteria



Each microenvironment, however, had its own characteristic microbiota.

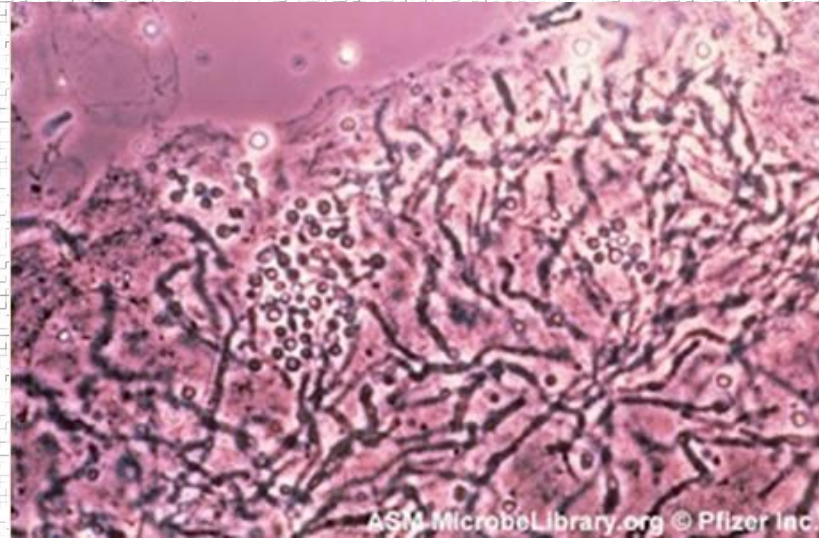
NMF of the SKIN: Other Bacteria

- ***Micrococcus***
- ***Neisseria* (non-pathogenic spp.)**
- **Diphtheroids**
- ***Streptococcus***
- ***Peptostreptococcus*** (NMF in gastrointestinal tract)
- ***Acinetobacter****
- ***Corynebacterium xerosis****

*** in lower []**

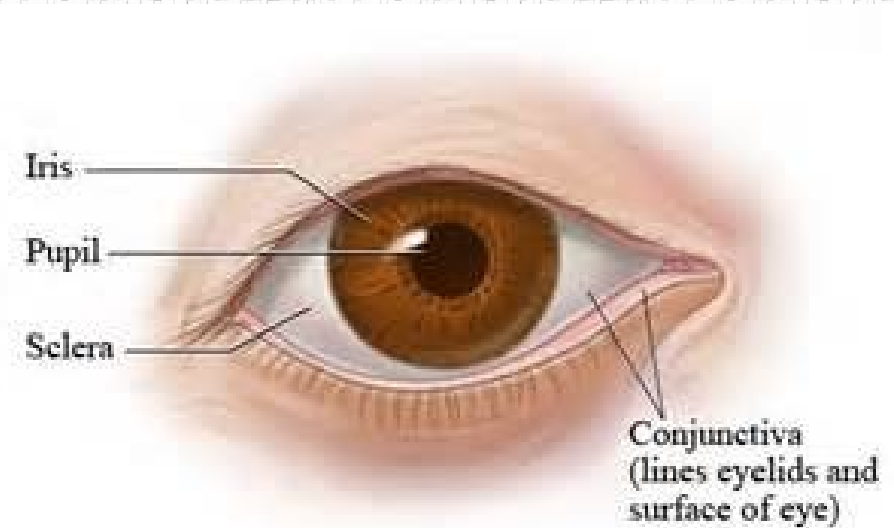
NMF of the SKIN: Fungi

- *Malassezia spp.*
– most common
- *Pityrosporum ovale* – lipophilic yeast found on scalp (dandruff)



NMF of the EYES (Conjunctiva): Bacteria

- ~ same microflora as the skin and MM
- tearflow controls the population ...
- ***Staphylococcus : epidermidis, aureus***
- **Enterobacteriaceae** (NMF in gastrointestinal tract)
- **Neisseriae**
- **Diphtheroids**
- **Non-hemolytic strep, etc.**



MOUTH (Oral Cavity)

- a very ideal environment for mcgs despite the presence of lysozyme in the saliva
- constant presence of food
- abundant moisture and warmth

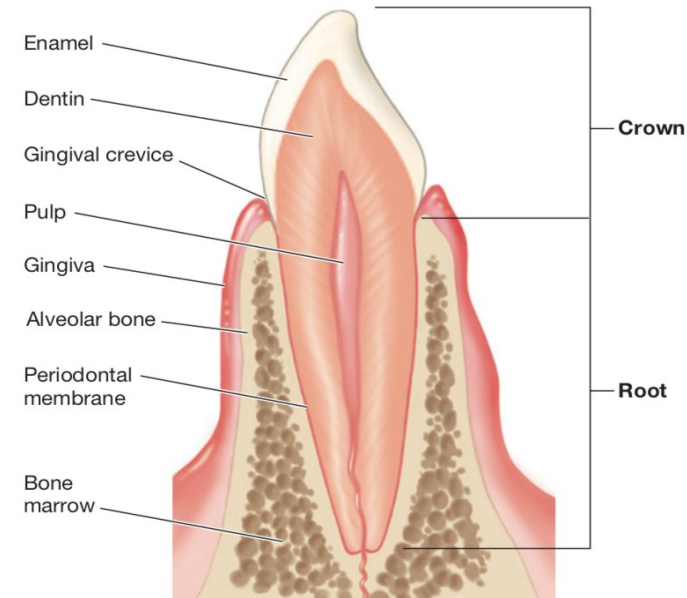


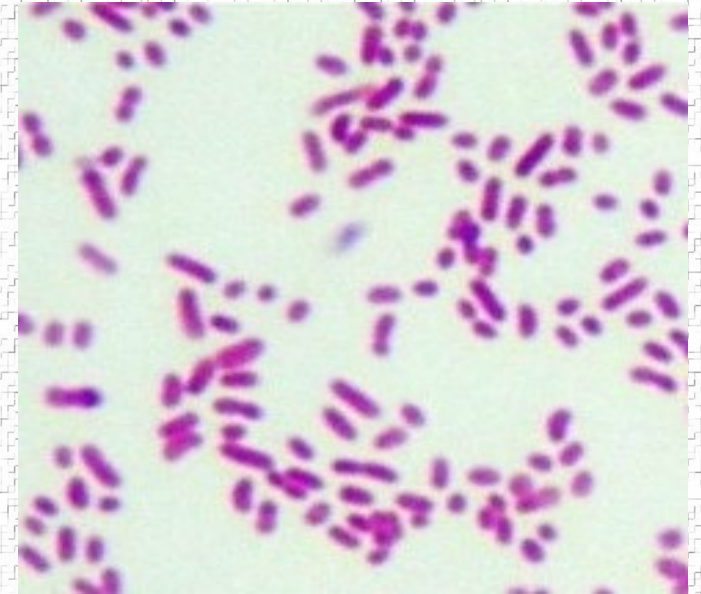
Figure 27.4 Section through a tooth. The diagram shows the tooth architecture and the surrounding tissues that anchor the tooth in the gum.

NMF of the MOUTH: Bacteria

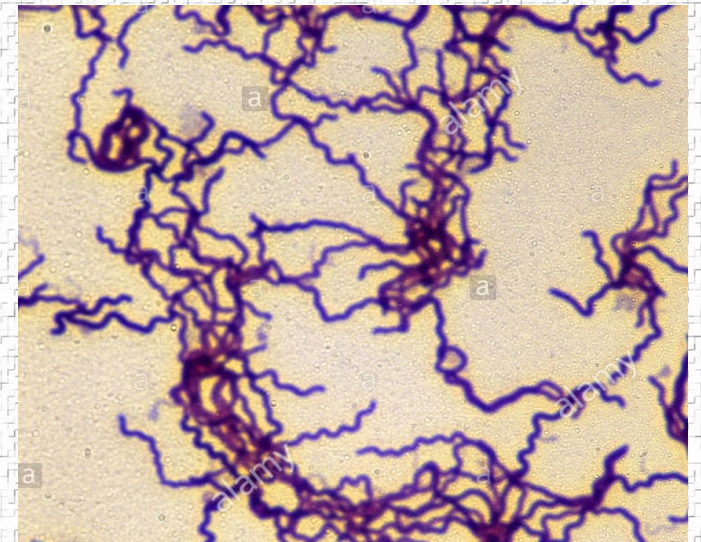
- *Prevotella* spp. – most prevalent
- *Treponema*
- *Actinomyces*
- *Bacteroides*
- *Capnocytophaga*
- *Lachnospiraceae*
- *Lactobacillus*
- *Leptotrichia*
- *Neisseria*
- *Selenomonas*
- *Streptococcus* spp. (esp. viridans streptococci)
- *Corynebacterium*
- *Fusobacterium**
- Gram-Neg. diplococci (*Moraxella*)
- Anaerobic spirochetes*

Others:

- *Candida*
- Protozoa



Prevotella



Treponema

Dental Caries and Mouth Microflora

- Some bacterial spp. (e.g. *S. mutans*) synthesize gelatinous deposits of glucans
 - Form a film on the teeth
 - Dental plaque
 - Other microorganisms attach
 - Fermenting species (e.g. lactobacilli and streptococci) produce acids
 - Destroy the teeth
 - contagious

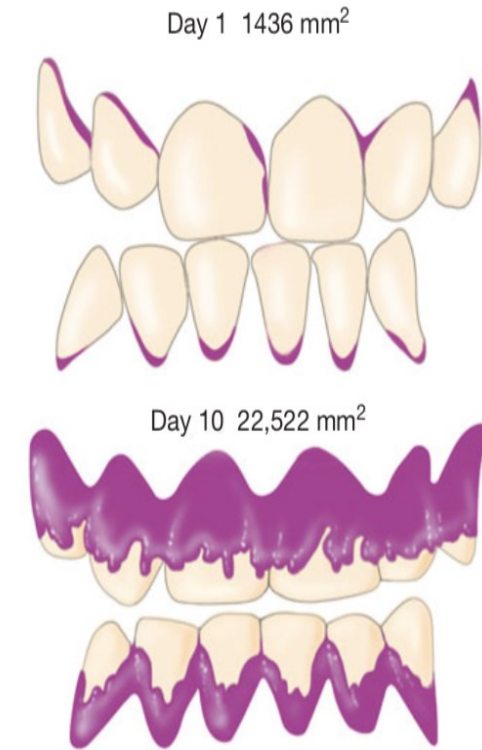


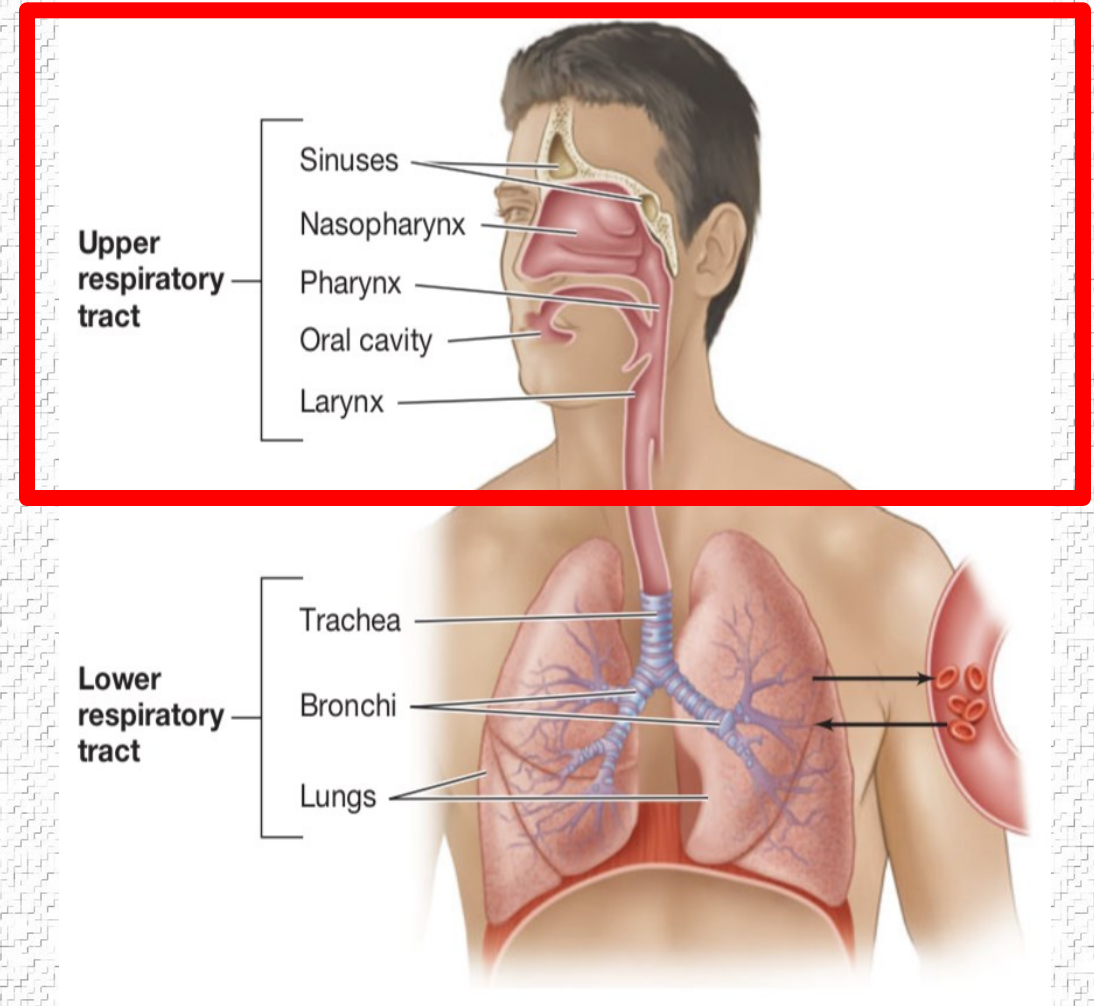
Figure 27.6 Distribution of dental plaque. Plaque is revealed by use of a disclosing agent on unbrushed teeth after 1 day (top) and 10 days (bottom). The stained areas indicate plaque. Plaque buildup starts near the gum line, beginning directly adjacent to the mucous membranes of the gingiva.

NMF of UPPER RESPIRATORY TRACT (URT)

➤ comprised by the sinuses, nasopharynx, pharynx, oral cavity and larynx

- have mechanical stratagem (e.g. sneezing, filters in nose)

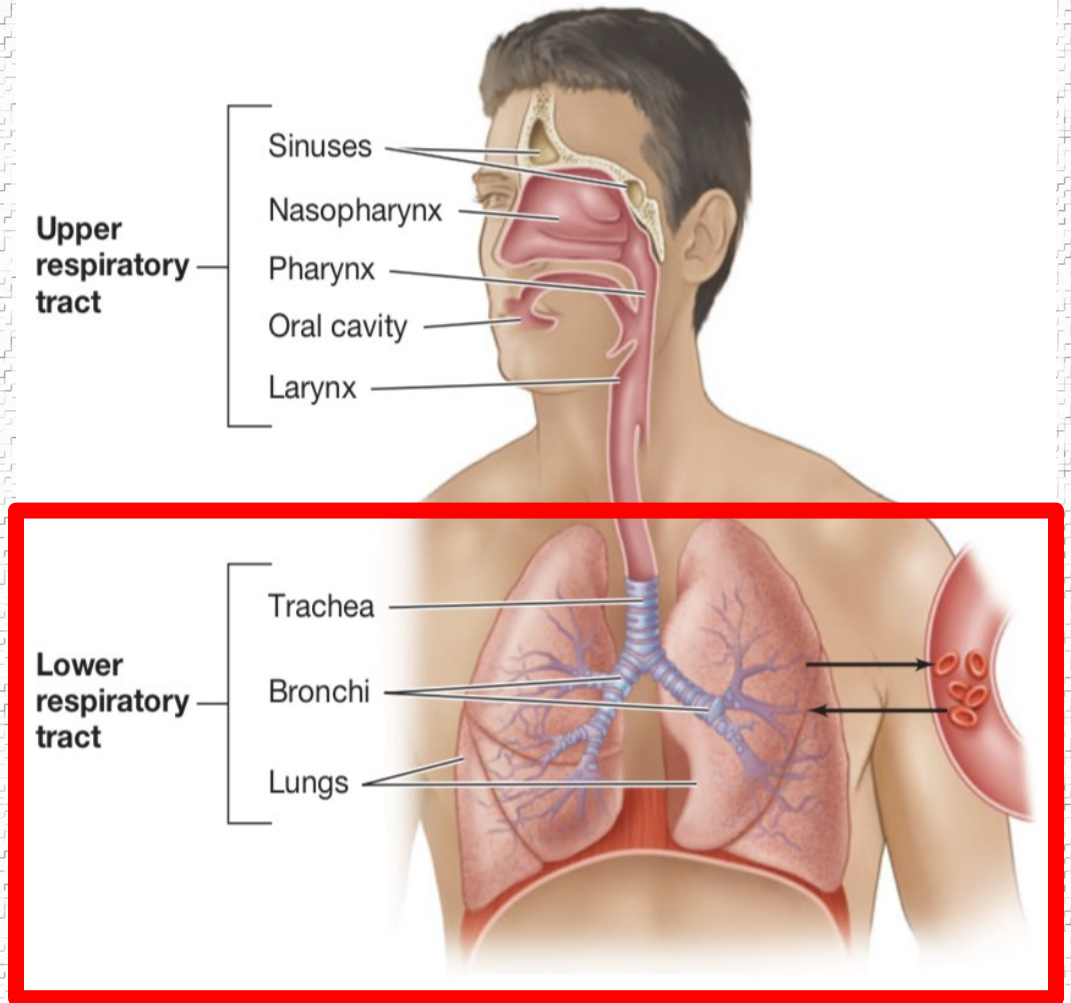
- **Nose :** *S. aureus*, *S. epidermidis*, *diphtheroids*
- **Throat :** as above + *S. pneumoniae*, viridans strep, *Haemophilus*, *Neisseria*
- **Pharynx :** non-hemolytic & α -hemolytic strep, *Neisseria*



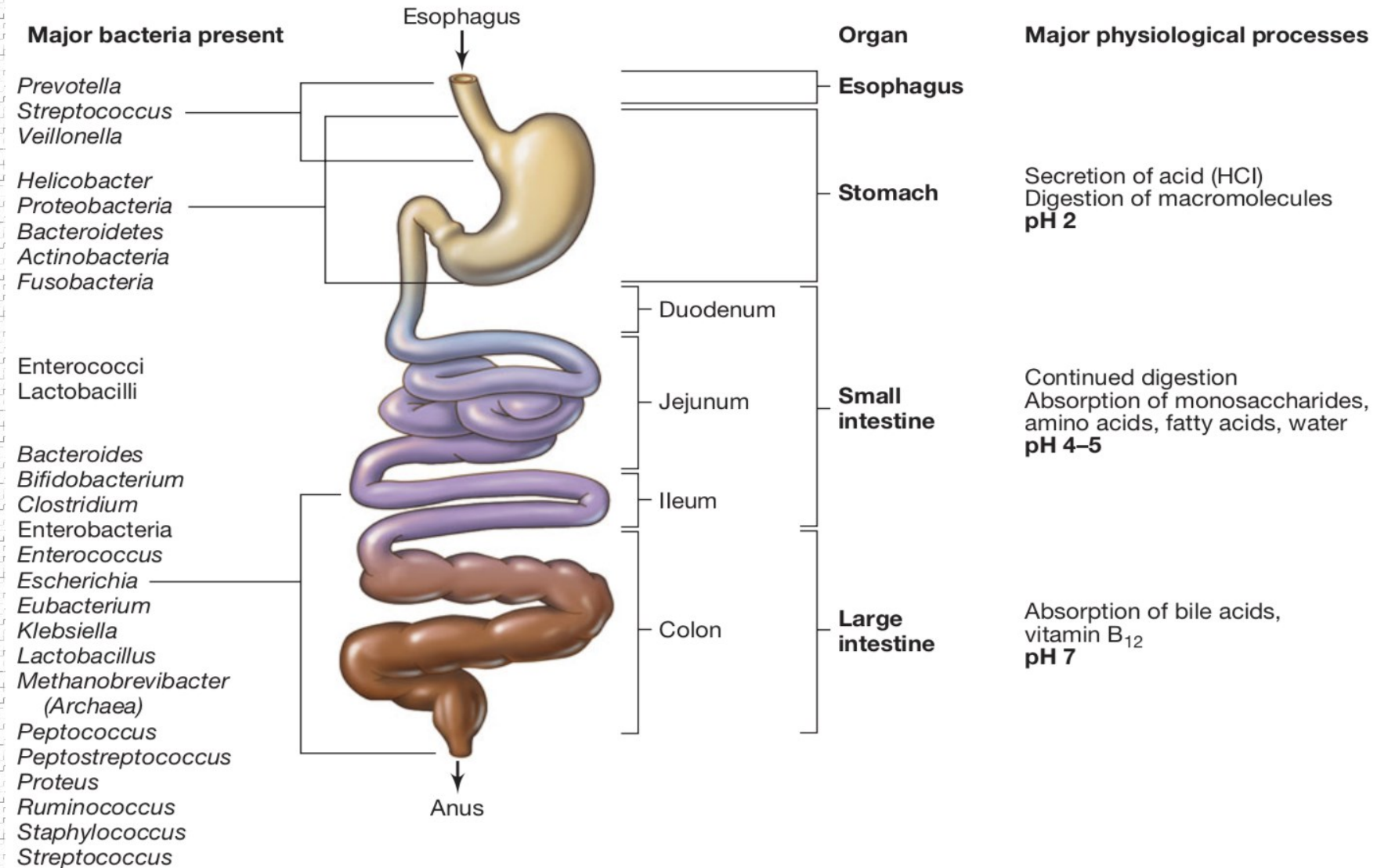
NMF of LOWER RESPIRATORY TRACT (LRT)

Comprised by the trachea, bronchi and lungs

- Has no resident microflora in healthy adults, despite the large number of organisms potentially able to reach this region during normal breathing.

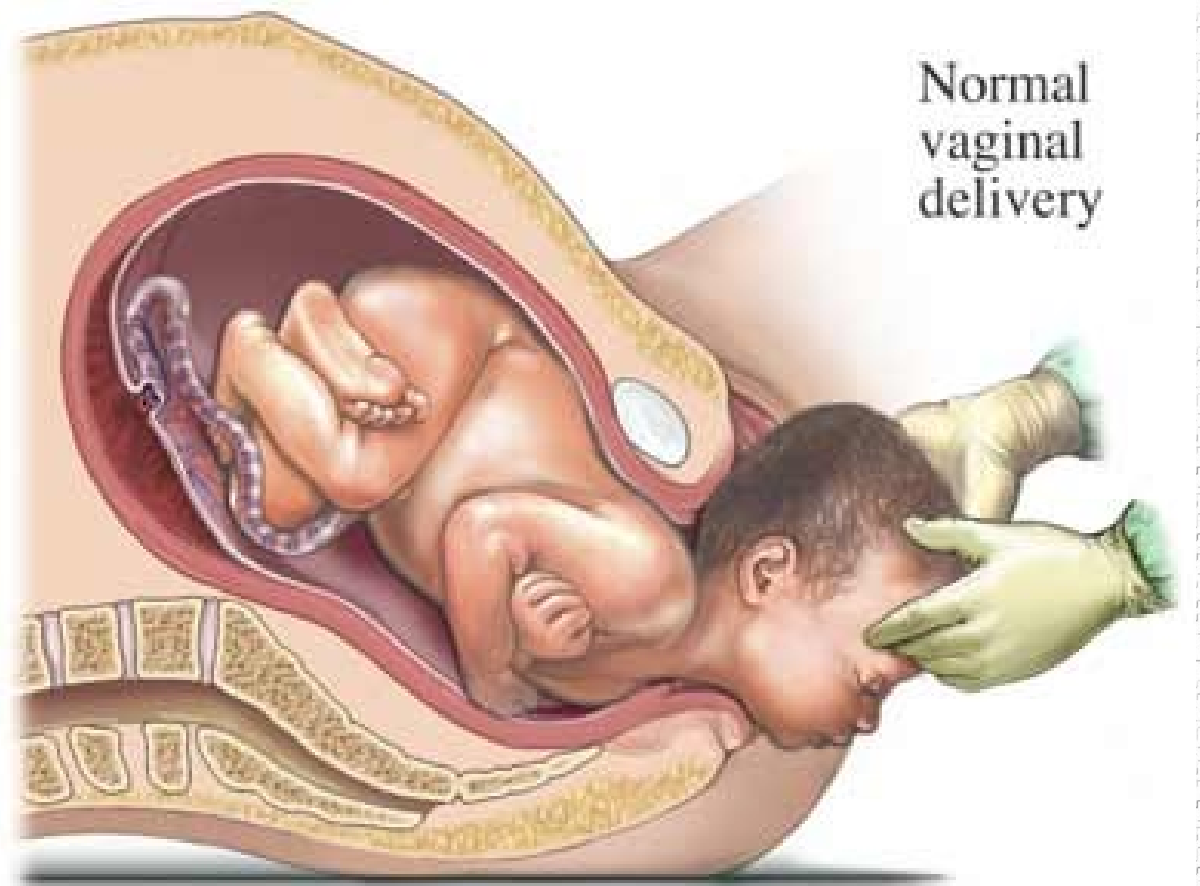


The Human GASTRO-INTESTINAL TRACT (GIT)



The Human GASTRO-INTESTINAL TRACT (GIT)

- Colonization occurs at birth
- Microbial population remains fairly stable unless exogenous factors disrupt the balance (e.g. antibiotic Tx).

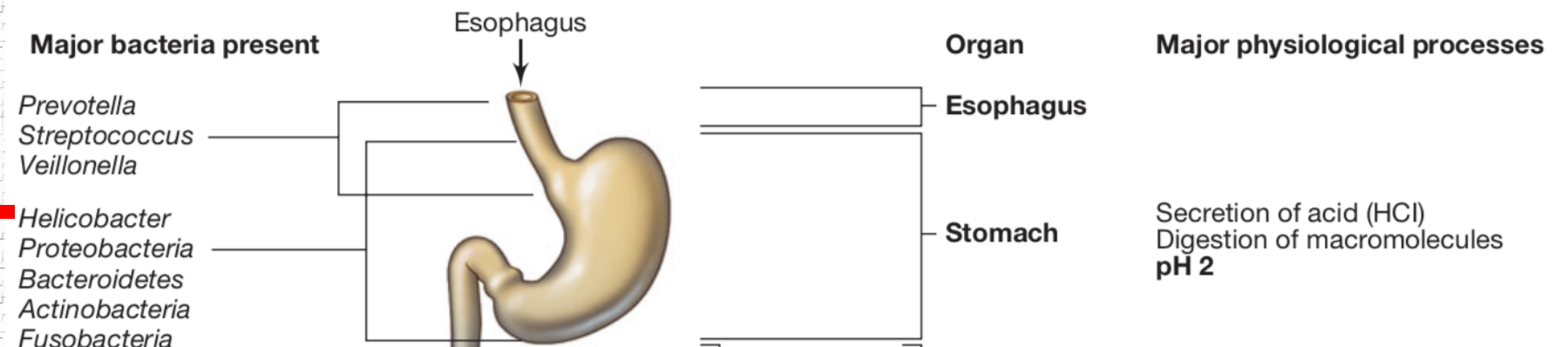


1. The Stomach

- has a natural acidity (~ pH 2) → a natural “microbial barrier”
 - regulates the population to a minimum (~ 10^3 – 10^5 /g)
 - protects against enteric pathogens

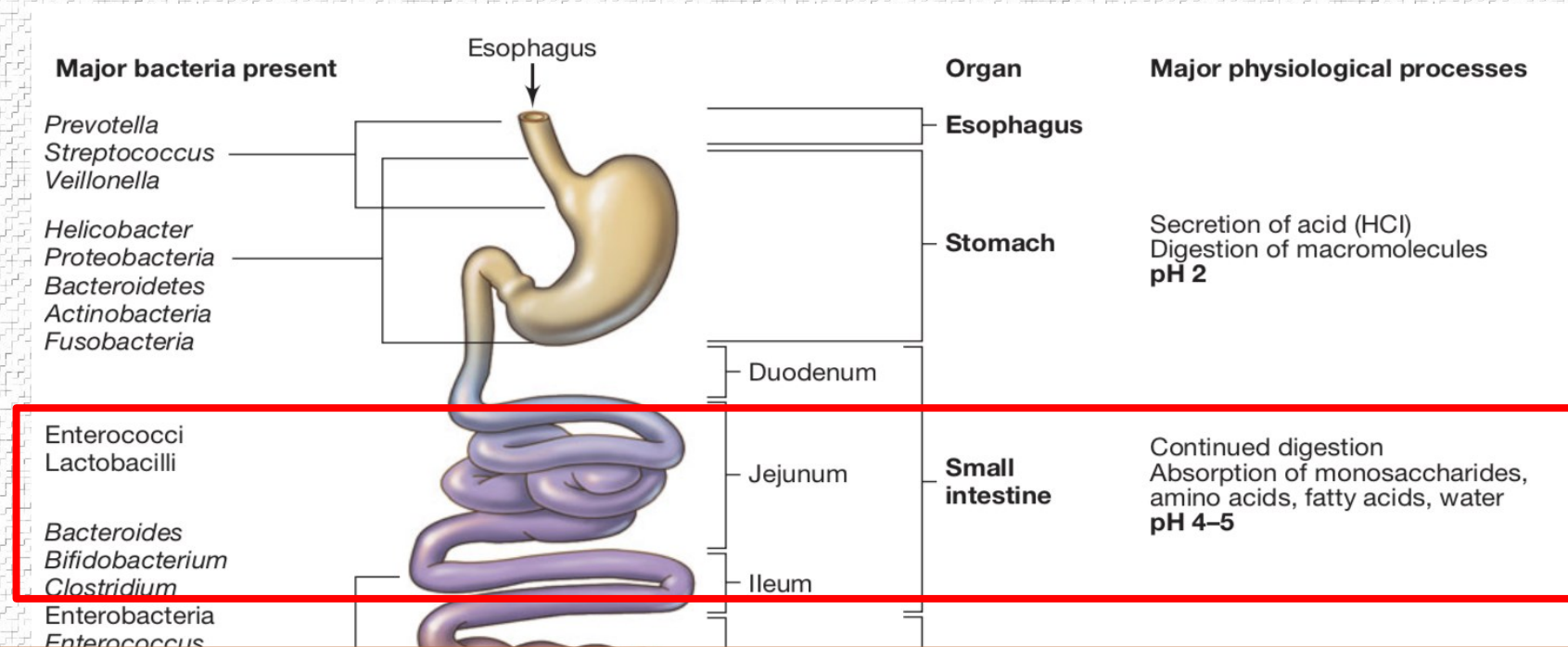


Helicobacter pylori - ulcer formation



2. The Small Intestine

Enterococci and Lactobacilli are facultative anaerobes
Bacteroides are anaerobes

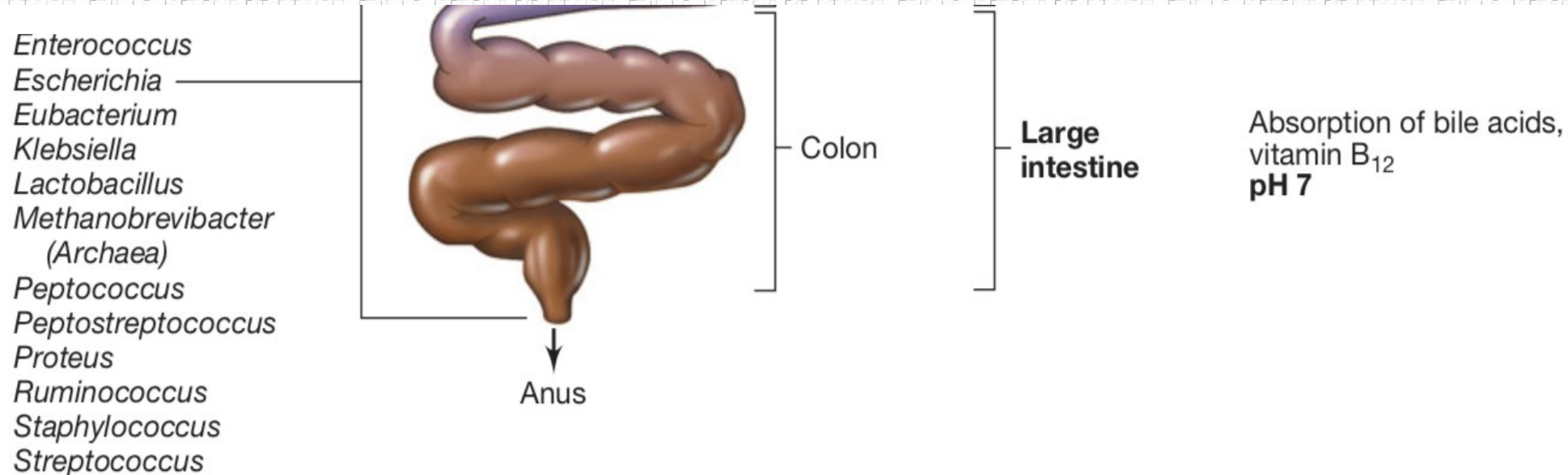


- Upper (duodenum): lactobacilli and enterococci ($\sim 10^3$ - 10^6 /g)
- Lower (jejunum/ileum): fecal microorganisms + low levels of *Salmonella** and *Campylobacter** ($\sim 10^5$ - 10^8 /g)

*asymptomatic residents

3. The Large Intestine

- mostly facultative anaerobes and aerobes, as it is close to anus
- ~ pH 7; like a “fermentation vessel”
- Has the largest number of resident flora (10^8 - 10^{10} , $> 10^{11}$ /gram feces) - 10-30% of fecal mass
- with ~ 100 distinct types of mcgs. present



Biochemical/metabolic contributions of intestinal microorganisms

<i>Process</i>	<i>Product</i>
Vitamin synthesis	Thiamine, riboflavin, pyridoxine, B ₁₂ , K
Gas production	CO ₂ , CH ₄ , H ₂
Odor production	H ₂ S, NH ₃ , amines, indole, skatole, butyric acid
Organic acid production	Acetic, propionic, butyric acids
Glycosidase reactions	β-Glucuronidase, β-galactosidase, β-glucosidase, α-glucosidase, α-galactosidase
Steroid metabolism (bile acids)	Esterified, dehydroxylated, oxidized, or reduced steroids

Factors Affecting the Normal Intestinal Flora

- antibiotic treatment → disruptive;
selection of drug-resistant forms of



staphylococci
Enterobacter, Proteus
Pseudomonas,
C. difficile, etc.

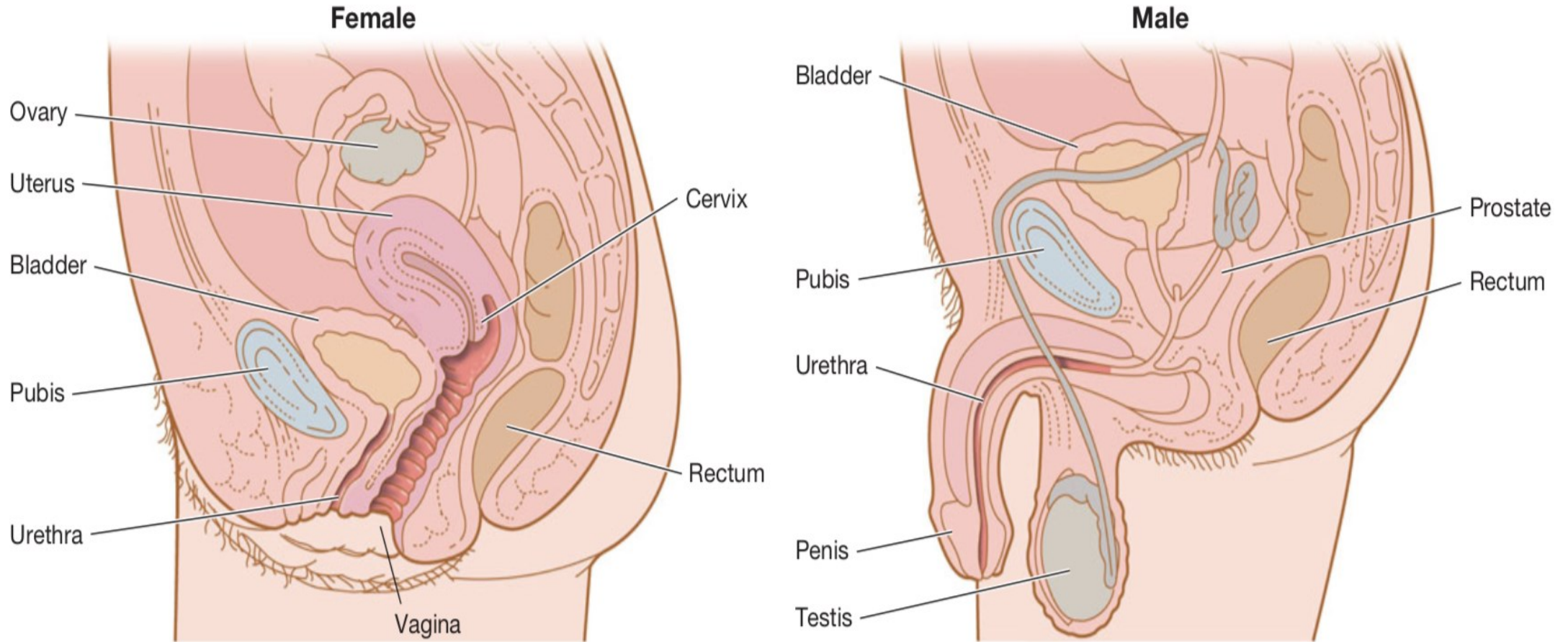
Factors Affecting the Normal Intestinal Flora

- **Exposure to other enteric pathogens**
(*Shigella*, EHEC, *Entamoeba histolytica*, etc.) (EHEC = enterohemorrhagic *Escherichia coli*)



also disruptive; may lead to disease

GENITOURINARY TRACT



In healthy male and female urogenital tracts, the kidneys and bladder itself are sterile, but the epithelial cells lining the distal urethra are colonized by facultatively aerobic gram-negative rods and cocci.

UROGENITAL TRACT

- Potential pathogens such as *E. coli* and *P. mirabilis*, normally present in small numbers in the body or local environment, can multiply in the urethra and become pathogenic under altered conditions such as changes in pH.
- Such organisms are a frequent cause of urinary tract infections, especially in women.
 - *E. coli* and *P. mirabilis* are opportunistic microorganisms.
 - *E. coli* causes UTI (urinary tract infection)

1. The Urethra

- predominant microflora are ~ avirulent



lactobacilli

← **corynebacteria**

coagulase (-) staphylococci

- * also facultatively anaerobic
Gram (-) rods (e.g. *E. coli* and *Proteus*) and cocci.

➤ **common agents of urethritis :**

Neisseria gonorrhoeae causes gonorrhea; gonococcal

Chlamydia trachomatis causes chlamydia; non gonococcal; most commonly detected in humans

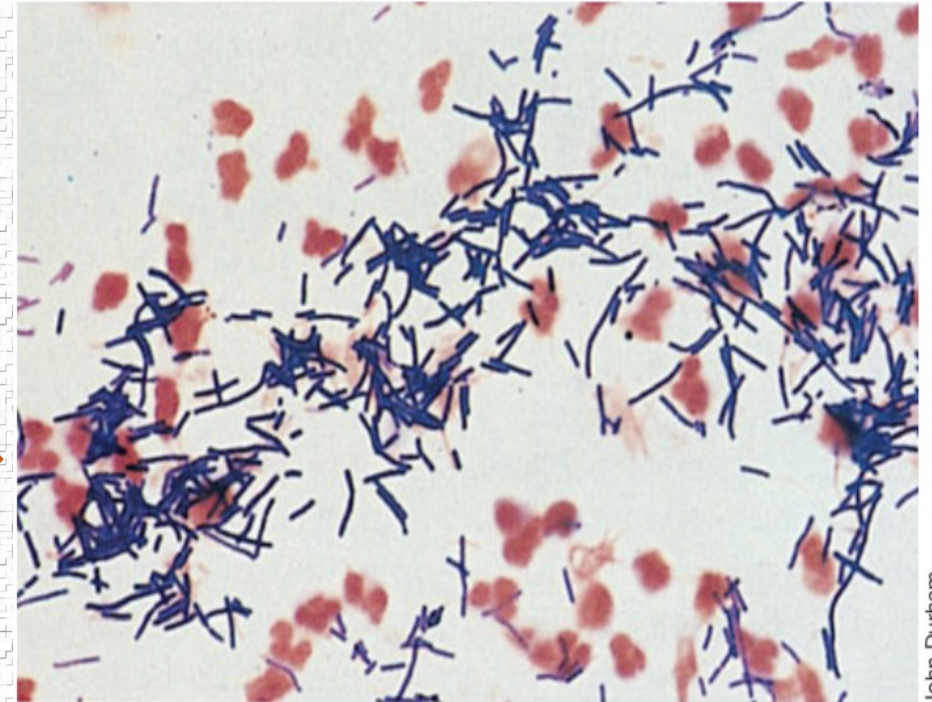


**isolation of any of these is always
clinically significant**

- Adults produce glycogen, which is used by lactobacillus to produce acid

2. The Vagina

- weakly acidic (in adults)
 - pH 5
- with significant amounts of glycogen → fermented → acidic products
- normal flora :
lactobacilli (*Lactobacillus acidophilus*, etc.), *E. coli*, streptococci, *Candida*, etc.



John Durham

Effect of age on the NMF :

- **Before puberty** → no glycogen; pH 7
(staph, strep, *E. coli*,
diphtheroids, etc.)
- **At puberty** → *L. acidophilus* ferments
(reprod. age) glycogen to produce acid;
pH 3.8 – 4.5
- **After menopause** → glycogen disappears;
pH rises (6-7) and flora is as in pre-puberty

Effect of hormonal factors (e.g. estrogen) on the NMF :

- At birth (high maternal estrogen) → ~lactobacilli
- After 6 weeks (maternal estrogen goes down)
- At puberty (estrogen increases) → lactobacilli increase in number and increase in glycogen
- After menopause (estrogen decreases)

Important causes of vaginitis (inflammation of the vagina):

1. ***Trichomonas vaginalis*** is a protozoa, not a bacteria
2. ***Candida glabrata***
3. ***Candida albicans***
 - type of yeast common in the vagina
 - not necessarily sexually transmitted
 - symptoms include vaginal itching, swelling, and irritation



➤ Common causes of genital disease

N. gonorrhoeae

Mycoplasma hominis

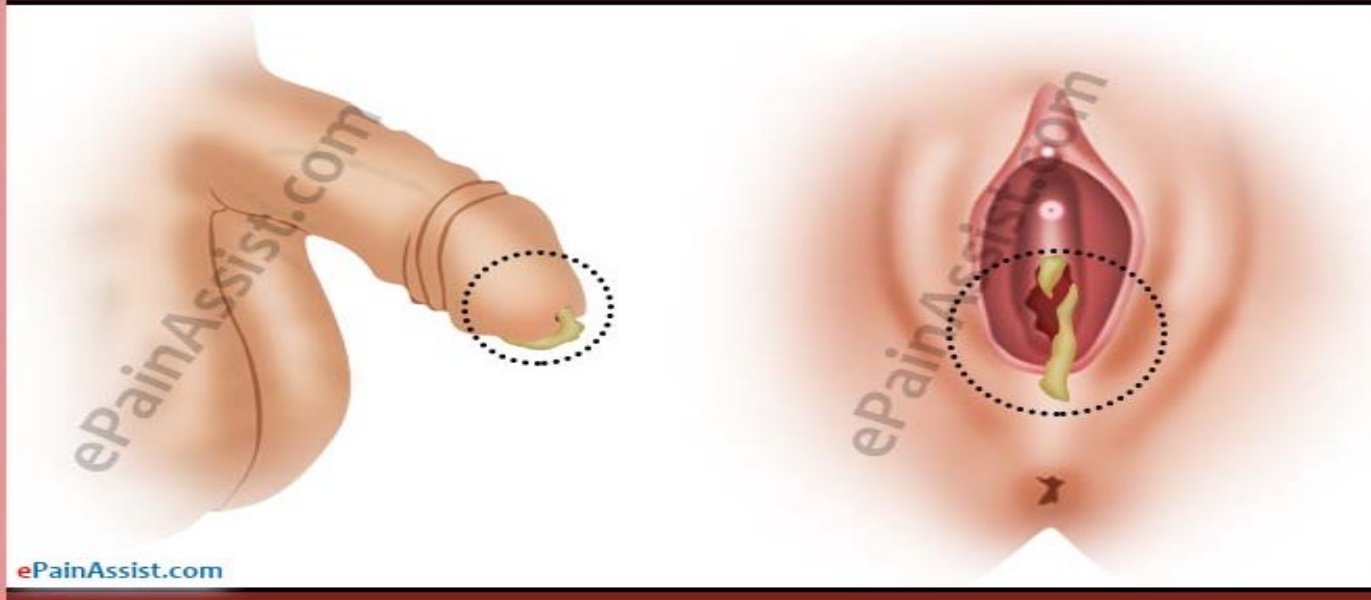
- has no cell wall; resistant to penicillin, ampicillin

C. trachomatis (Tulo)

- occurs in male and female; symptoms appear weeks after intimacy

- Chlamydia Trachomatis is one of the most common STD- sexually transmitted disease or infection which is completely curable.
- It is a sexually transmitted disease which spreads through unprotected anal, vaginal or oral sex with a person infected with Chlamydia.

For More Information:
Visit: www.epainassist.com



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